

LAB EXPERIMENTS 11-15

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SET 11: Product Category Analysis

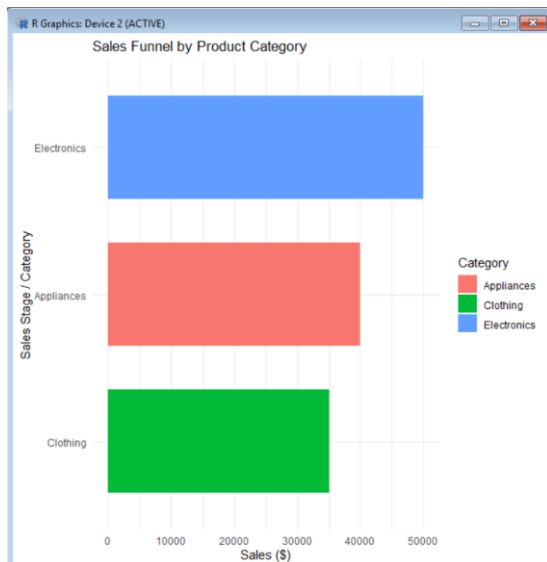
Q1. Funnel Chart – Sales Conversion Process

```
library(ggplot2)

sales <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

sales <- sales[order(-sales$Sales), ]

ggplot(sales,
  aes(x = reorder(Category, Sales),
    y = Sales,
    fill = Category)) +
  geom_bar(stat = "identity",
    width = 0.7) +
  coord_flip() +
  labs(
    title = "Sales Funnel by Product Category",
    x = "Sales Stage / Category",
    y = "Sales ($)"
  ) +
  theme_minimal()
```



Q2. Sales Data Table

```
sales <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
cat("Product Category Sales Data\n\n")
print(sales)
```

```
> # Display Table
> cat("Product Category Sales Data\n\n")
Product Category Sales Data

>
> print(sales)
  Category Sales
1 Electronics 50000
2  Clothing 35000
3 Appliances 40000
> |
```

Q3. R Shiny Dashboard

```
library(shiny)
library(ggplot2)
sales <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
```

```

)
ui <- fluidPage(
  titlePanel("Product Category Sales Dashboard"),
  mainPanel(
    plotOutput("pieChart"),
    plotOutput("funnelChart"),
    tableOutput("salesTable")
  )
)
server <- function(input, output) {
  output$pieChart <- renderPlot({
    ggplot(sales,
      aes(x = "",
          y = Sales,
          fill = Category)) +
    geom_bar(stat = "identity",
      width = 1) +
    coord_polar("y") +
    geom_text(
      aes(label = paste(Category, Sales)),
      position = position_stack(vjust = 0.5)
    ) +
    labs(title = "Sales Distribution") +
    theme_void()
  })
  output$funnelChart <- renderPlot({
    ggplot(sales,
      aes(x = reorder(Category, Sales),

```

```

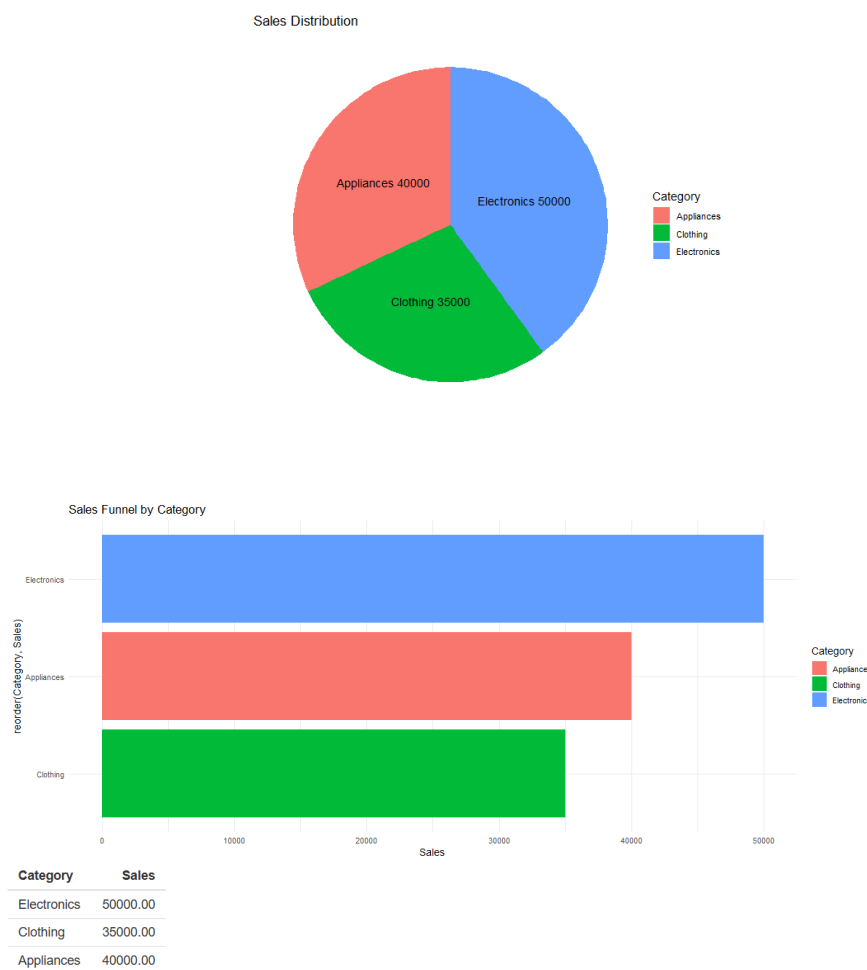
    y = Sales,
    fill = Category)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(title = "Sales Funnel by Category") +
  theme_minimal()
})

output$salesTable <- renderTable({
  sales
})
}

shinyApp(ui = ui, server = server)

```

Product Category Sales Dashboard



Q4. Pie Chart – Sales Distribution

```
library(ggplot2)

sales <- data.frame(

  Category = c("Electronics", "Clothing", "Appliances"),

  Sales = c(50000, 35000, 40000)

)

ggplot(sales,

  aes(x = "",

    y = Sales,

    fill = Category)) +

  geom_bar(stat = "identity",

    width = 1) +

  coord_polar("y") +

  geom_text(

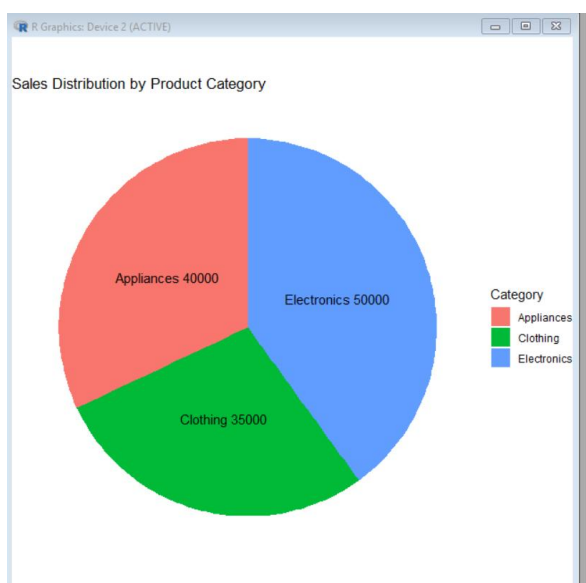
    aes(label = paste(Category, Sales)),

    position = position_stack(vjust = 0.5)

  ) +

  labs(title = "Sales Distribution by Product Category") +

  theme_void()
```



SET 12: Website Traffic

Q1. Line Chart – Daily Page Views

```
library(ggplot2)

traffic <- data.frame(

  Date = as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03")),

  PageViews = c(1500, 1600, 1400),

  CTR = c(2.3, 2.7, 2.0)

)

ggplot(traffic,

  aes(Date, PageViews, group = 1)) +

  geom_line(linewidth = 1.2) +

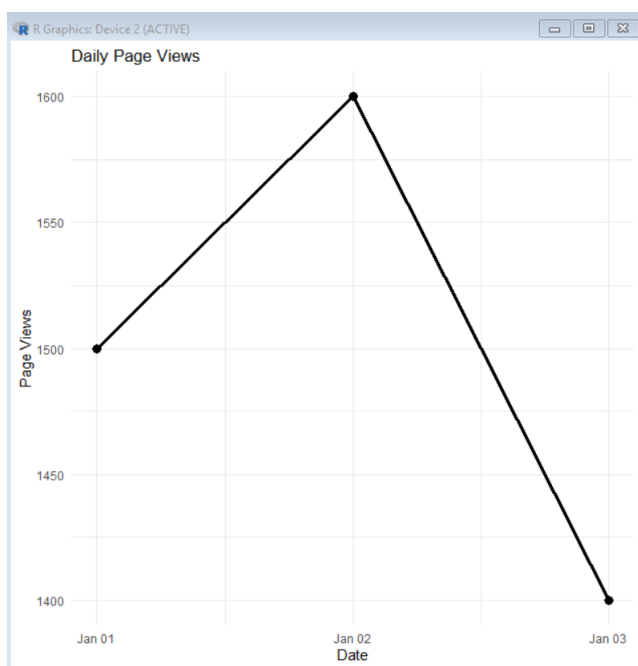
  geom_point(size = 3) +

  labs(

    title = "Daily Page Views",

    x = "Date",

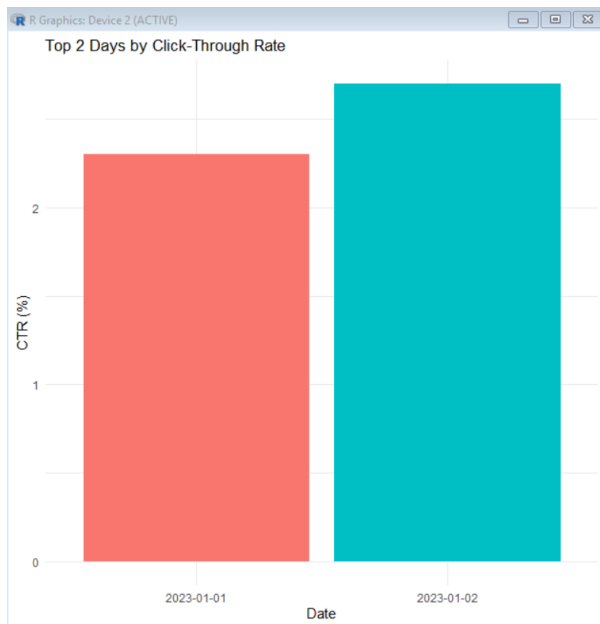
    y = "Page Views"
```



```
) +  
theme_minimal()
```

Q2. Bar Chart – Top N Days by CTR

```
library(ggplot2)  
  
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01",  
                   "2023-01-02",  
                   "2023-01-03")),  
  PageViews = c(1500, 1600, 1400),  
  CTR = c(2.3, 2.7, 2.0)  
)  
  
top_n <- head(  
  traffic[order(-traffic$CTR), ],  
  2  
)  
  
ggplot(top_n,  
  aes(x = reorder(Date, CTR),  
      y = CTR,  
      fill = factor(Date))) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Top 2 Days by Click-Through Rate",  
    x = "Date",  
    y = "CTR (%)"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```



Q3. Website Traffic Table

Dataset

```
traffic <- data.frame(  
  Date = c("2023-01-01",  
            "2023-01-02",  
            "2023-01-03"),  
  Page_Views = c(1500, 1600, 1400),  
  Click_Through_Rate = c("2.3%", "2.7%", "2.0%")  
)  
cat("Website Traffic Data\n\n")  
print(traffic)  
  
>  
> print(traffic)  
      Date Page_Views Click_Through_Rate  
1 2023-01-01      1500             2.3%  
2 2023-01-02      1600             2.7%  
3 2023-01-03      1400             2.0%  
> |
```

Q4. R Shiny Dashboard

```
library(shiny)
```

```
library(ggplot2)
```

```

traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
                    "2023-01-02",
                    "2023-01-03")),
  PageViews = c(1500, 1600, 1400),
  CTR = c(2.3, 2.7, 2.0)
)

ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  mainPanel(
    plotOutput("lineChart"),
    plotOutput("barChart"),
    tableOutput("trafficTable")
  )
)

server <- function(input, output) {
  output$lineChart <- renderPlot({

    ggplot(traffic,
      aes(Date, PageViews, group = 1)) +
    geom_line(linewidth = 1.2) +
    geom_point(size = 3) +
    labs(title = "Daily Page Views") +
    theme_minimal()
  })

  output$barChart <- renderPlot({
    ggplot(traffic,
      aes(Date, CTR)) +

```

```

geom_bar(stat = "identity") +
labs(
  title = "Click-Through Rate by Day",
  x = "Date",
  y = "CTR (%)"
) +
theme_minimal()
})

```

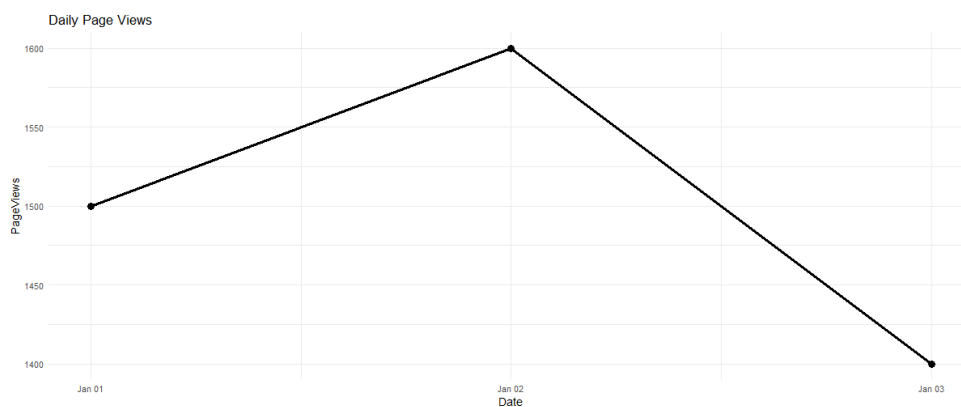
```

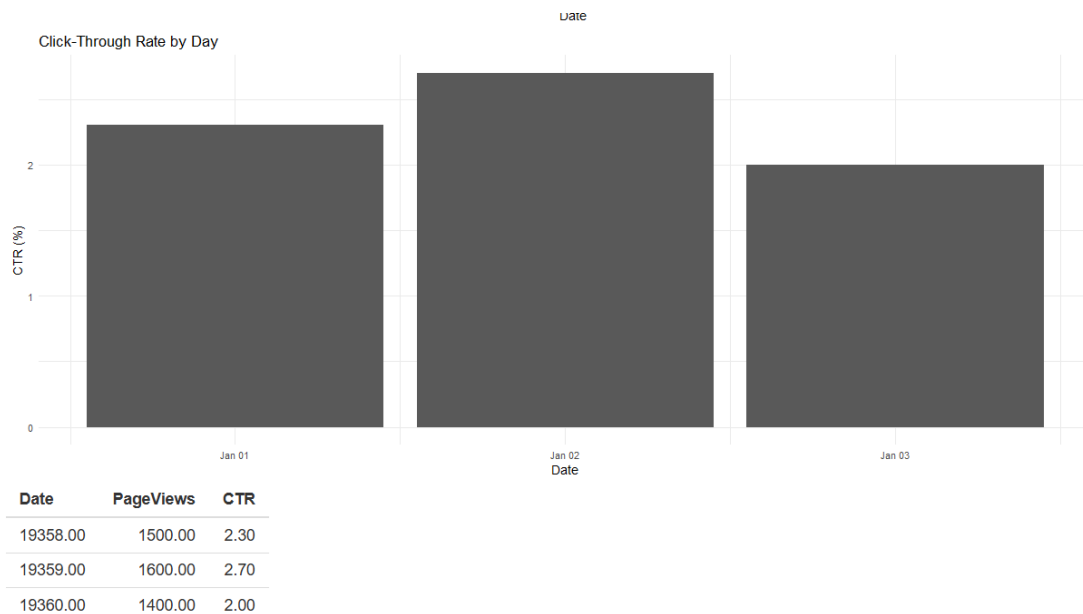
output$trafficTable <- renderTable({
  traffic
})
}

```

```
shinyApp(ui = ui, server = server)
```

Website Traffic Dashboard





SET 13: Geographic Data

Q1. Map Chart

Since the dataset gives city names but not latitude and longitude, we add approximate coordinates for visualization.

```
library(ggplot2)
```

```
geo <- data.frame(
```

```
  City = c("City A", "City B", "City C"),
```

```
  Population = c(500000, 700000, 600000),
```

```
  Temperature = c(75, 68, 80),
```

```
  Elevation = c(1000, 800, 1200),
```

```
  Longitude = c(80.27, 77.59, 78.48),
```

```
  Latitude = c(13.08, 12.97, 17.38)
```

```
)
```

```
ggplot(geo,
```

```
  aes(Longitude,
```

```
    Latitude,
```

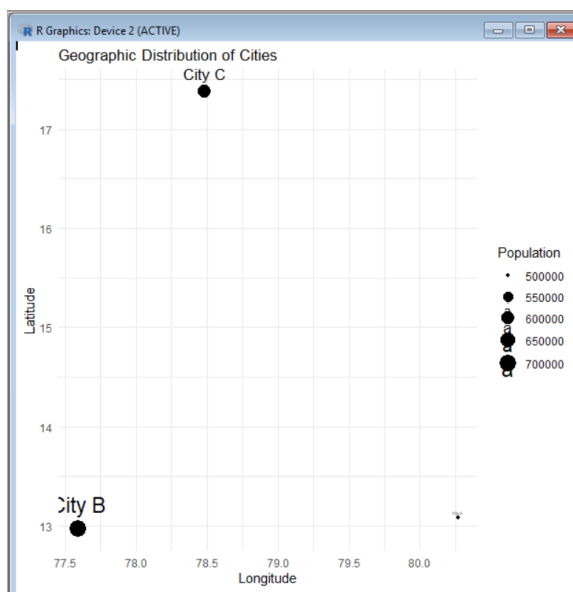
```
    size = Population,
```

```
    label = City)) +
```

```

geom_point() +
geom_text(vjust = -1) +
labs(
  title = "Geographic Distribution of Cities",
  x = "Longitude",
  y = "Latitude"
) +
theme_minimal()

```



Q2. Scatter Plot – Temperature vs Population

```

library(ggplot2)

geo <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80)
)

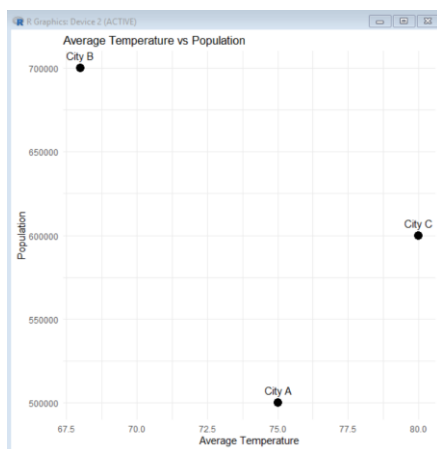
ggplot(geo,
  aes(Temperature,
    Population,
    label = City)) +

```

```

geom_point(size = 4) +
geom_text(vjust = -1) +
labs(
  title = "Average Temperature vs Population",
  x = "Average Temperature",
  y = "Population"
) +
theme_minimal()

```



Insight

- City B has the highest population but the lowest average temperature.
- City C has the highest temperature and a medium-sized population.
- With only three cities, no strong relationship can be confirmed.

Q3. Geographic Data Table

Dataset

```

geo <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Average_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)

```

```
Cat("Geographic Data\n\n")
```

```
print(geo)
```

```
> print(geo)
  City Population Average_Temperature Elevation
1 City A      5e+05                75      1000
2 City B      7e+05                68        800
3 City C      6e+05                80      1200
> |
```

Q4. R Shiny Dashboard

```
library(shiny)
```

```
library(ggplot2)
```

```
geo <- data.frame(
```

```
  City = c("City A", "City B", "City C"),
```

```
  Population = c(500000, 700000, 600000),
```

```
  Temperature = c(75, 68, 80),
```

```
  Elevation = c(1000, 800, 1200),
```

```
  Longitude = c(80.27, 77.59, 78.48),
```

```
  Latitude = c(13.08, 12.97, 17.38)
```

```
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Geographic Data Dashboard"),
```

```
  mainPanel(
```

```
    plotOutput("map"),
```

```
    plotOutput("scatter"),
```

```
    tableOutput("geoTable")
```

```
)
```

```
)
```

```
server <- function(input, output) {
```

```
  output$map <- renderPlot({
```

```
    ggplot(geo,
```

```

    aes(Longitude,
        Latitude,
        size = Population,
        label = City)) +
  geom_point() +
  geom_text(vjust = -1) +
  labs(title = "City Geographic Distribution") +
  theme_minimal()
})

```

```

output$scatter <- renderPlot({

```

```

  ggplot(geo,
    aes(Temperature,
        Population,
        label = City)) +
  geom_point(size = 4) +
  geom_text(vjust = -1) +
  labs(
    title = "Temperature vs Population",
    x = "Average Temperature",
    y = "Population"
  ) +
  theme_minimal()
})

```

```

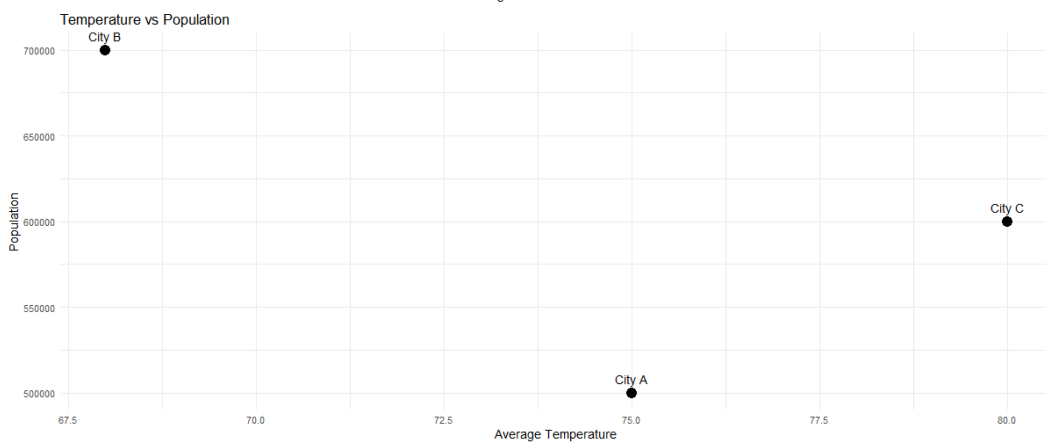
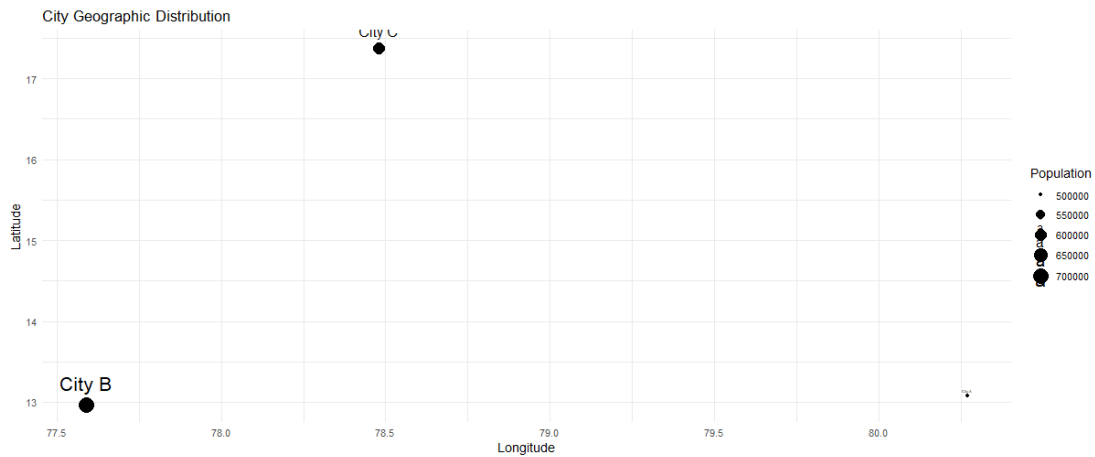
output$geoTable <- renderTable({
  geo
})

```

```
}
```

```
shinyApp(ui = ui, server = server)
```

Geographic Data Dashboard



City	Population	Temperature	Elevation	Longitude	Latitude
City A	500000.00	75.00	1000.00	80.27	13.08
City B	700000.00	68.00	800.00	77.59	12.97
City C	600000.00	80.00	1200.00	78.48	17.38

SET 14: Survey Results

Q1. Stacked Bar Chart – Question 1

```
library(ggplot2)
```

```
survey <- data.frame(
```

```
  Respondent = c(1, 2, 3),
```

```
  Question1 = c("A", "B", "C"),
```

```
  Question2 = c("B", "A", "A"),
```

```
  Question3 = c("C", "D", "B")
```

```
)
```

```
# Stacked Bar Chart
```

```
ggplot(survey,  
       aes(x = "Question 1",  
           fill = Question1)) +  
geom_bar() +  
labs(  
  title = "Distribution of Question 1 Answers",  
  x = "Question",  
  y = "Number of Responses",  
  fill = "Answer"  
) +  
theme_minimal()
```

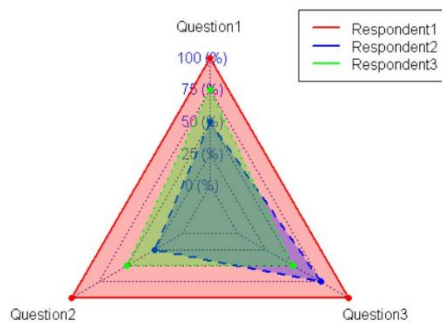


Q2. Radar Chart – Overall Response Pattern

```
library(fmsb)  
  
survey <- data.frame(  
  Respondent = c(1, 2, 3),  
  Question1 = c("A", "B", "C"),  
  Question2 = c("B", "A", "A"),  
  Question3 = c("C", "D", "B")
```

```
)
responses <- c(
  A = sum(survey$Question1 == "A") +
    sum(survey$Question2 == "A") +
    sum(survey$Question3 == "A"),
  B = sum(survey$Question1 == "B") +
    sum(survey$Question2 == "B") +
    sum(survey$Question3 == "B"),
  C = sum(survey$Question1 == "C") +
    sum(survey$Question2 == "C") +
    sum(survey$Question3 == "C"),
  D = sum(survey$Question1 == "D") +
    sum(survey$Question2 == "D") +
    sum(survey$Question3 == "D")
)
```

```
radar_data <- data.frame(
  Maximum = rep(max(responses), 4),
  Minimum = rep(0, 4),
  Responses = responses
)
rownames(radar_data) <- names(responses)
radarchart(
  radar_data,
  title = "Overall Survey Response Pattern"
)
```



Q3. Survey Response Table

```
survey <- data.frame(
  Respondent = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

cat("Survey Results\n\n")

print(survey)
```

```
> print(survey)
  Respondent Question_1 Question_2 Question_3
1          1          A          B          C
2          2          B          A          D
3          3          C          A          B
> |
```

Q4. R Shiny Dashboard

```
library(shiny)

library(ggplot2)

survey <- data.frame(
  Respondent = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
```

```

Question2 = c("B", "A", "A"),
Question3 = c("C", "D", "B")
)

ui <- fluidPage(
  titlePanel("Survey Results Dashboard"),
  mainPanel(
    plotOutput("stackedBar"),
    plotOutput("radar"),
    tableOutput("surveyTable")
  )
)

server <- function(input, output) {
  output$stackedBar <- renderPlot({
    ggplot(survey,
      aes(x = "Question 1",
        fill = Question1)) +
    geom_bar() +
    labs(
      title = "Question 1 Answer Distribution",
      x = "Question",
      y = "Responses"
    ) +
    theme_minimal()
  })

  output$radar <- renderPlot({
    responses <- c(
      A = sum(survey$Question1 == "A") +

```

```

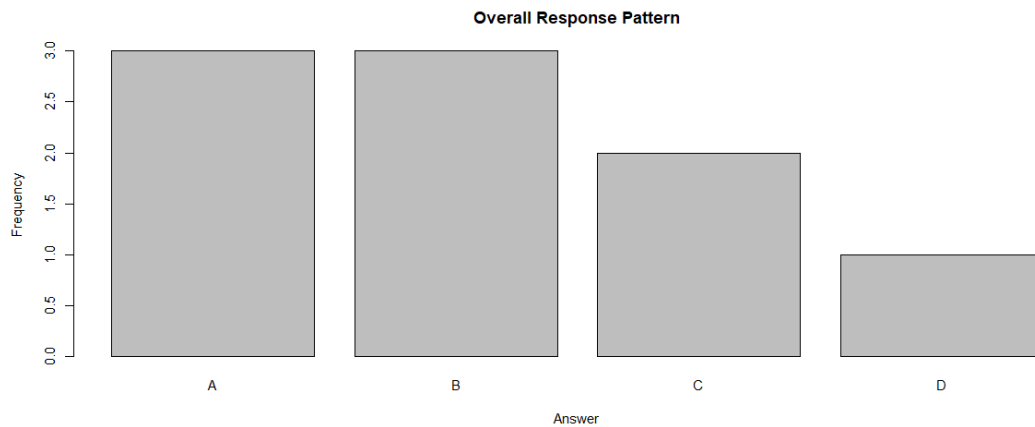
    sum(survey$Question2 == "A") +
    sum(survey$Question3 == "A"),

    B = sum(survey$Question1 == "B") +
    sum(survey$Question2 == "B") +
    sum(survey$Question3 == "B"),
    C = sum(survey$Question1 == "C") +
    sum(survey$Question2 == "C") +
    sum(survey$Question3 == "C"),
    D = sum(survey$Question1 == "D") +
    sum(survey$Question2 == "D") +
    sum(survey$Question3 == "D")
  )
  barplot(
    responses,
    main = "Overall Response Pattern",
    xlab = "Answer",
    ylab = "Frequency"
  )
})
output$surveyTable <- renderTable({
  survey
})
}

shinyApp(ui = ui, server = server)

```

Survey Results Dashboard



Respondent	Question1	Question2	Question3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

SET 15: Product Sales Analysis

Q1. Grouped Bar Chart

```
library(ggplot2)
```

```
library(tidyr)
```

```
sales <- data.frame(
```

```
  ProductID = c(1, 2, 3),
```

```
  ProductName = c("Product A", "Product B", "Product C"),
```

```
  January = c(2000, 1500, 1200),
```

```
  February = c(2200, 1800, 1400),
```

```
  March = c(2400, 1600, 1100)
```

```
)
```

```
sales_long <- pivot_longer(
```

```

sales,
cols = c(January, February, March),
names_to = "Month",
values_to = "Sales"
)

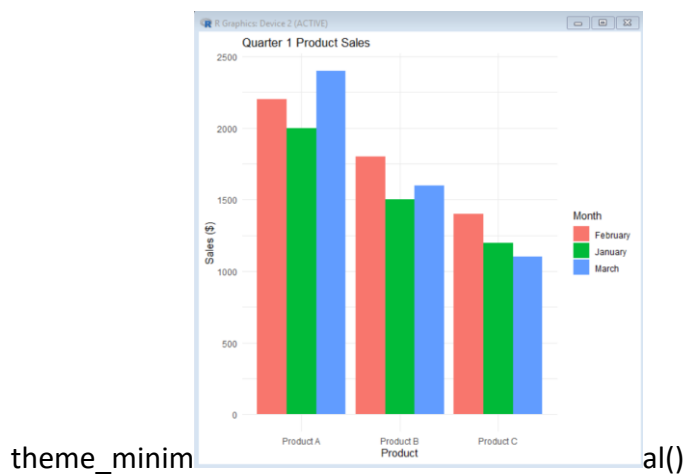
```

Grouped Bar Chart

```

ggplot(sales_long,
  aes(ProductName,
    Sales,
    fill = Month)) +
geom_bar(
  stat = "identity",
  position = "dodge"
) +
labs(
  title = "Quarter 1 Product Sales",
  x = "Product",
  y = "Sales ($)"
) +

```



theme_minimal()

Q2. Stacked Area Chart

```
library(ggplot2)

library(tidyr)

sales <- data.frame(
  ProductName = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

sales_long <- pivot_longer(
  sales,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)

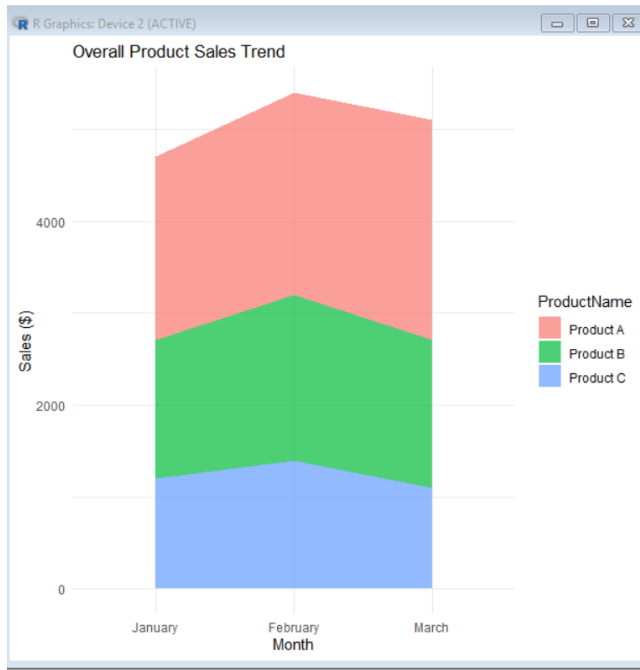
sales_long$Month <- factor(
  sales_long$Month,
  levels = c("January", "February", "March")
)

ggplot(sales_long,
  aes(Month,
    Sales,
    fill = ProductName,
    group = ProductName)) +
  geom_area(alpha = 0.7) +
  labs(
    title = "Overall Product Sales Trend",
```

```

x = "Month",
y = "Sales ($)"
) +
theme_minimal()

```



Q3. Monthly Sales Table

```

sales <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)
cat("Monthly Product Sales\n\n")
print(sales)

```

```
> print(sales)
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A         2000         2200         2400
2          2   Product B         1500         1800         1600
3          3   Product C         1200         1400         1100
> |
```

Q4. R Shiny Dashboard

```
library(shiny)
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
sales <- data.frame(
```

```
  ProductName = c("Product A", "Product B", "Product C"),
```

```
  January = c(2000, 1500, 1200),
```

```
  February = c(2200, 1800, 1400),
```

```
  March = c(2400, 1600, 1100)
```

```
)
```

```
sales_long <- pivot_longer(
```

```
  sales,
```

```
  cols = c(January, February, March),
```

```
  names_to = "Month",
```

```
  values_to = "Sales"
```

```
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Monthly Product Sales Dashboard"),
```

```
  sidebarLayout(
```

```
    sidebarPanel(
```

```
      selectInput(
```

```
        "product",
```

```
        "Select Product",
```

```
        choices = c(
```

```

      "All",
      unique(sales_long$ProductName)
    )
  )
),
mainPanel(
  plotOutput("groupedBar"),

  plotOutput("areaChart"),

  tableOutput("salesTable")

)
)
)
server <- function(input, output) {
  filtered <- reactive({
    if (input$product == "All") {
      sales_long
    } else {
      subset(
        sales_long,
        ProductName == input$product
      )
    }
  })
  output$groupedBar <- renderPlot({
    ggplot(

```

```

    filtered(),
    aes(
      ProductName,
      Sales,
      fill = Month
    )
  ) +
  geom_bar(
    stat = "identity",
    position = "dodge"
  ) +
  labs(
    title = "Grouped Product Sales"
  ) +
  theme_minimal()
})

output$areaChart <- renderPlot({
  ggplot(
    filtered(),
    aes(
      Month,
      Sales,
      fill = ProductName,
      group = ProductName
    )
  ) +
  geom_area(alpha = 0.7) +
  labs(

```

```

    title = "Stacked Area Sales Trend"
  ) +
  theme_minimal()
})

```

```

output$salesTable <- renderTable({

```

```

  if (input$product == "All") {
    sales
  } else {
    subset(
      sales,
      ProductName == input$product
    )
  }
})
}

shinyApp(ui = ui, server = server)

```

